

Problem Set #1: 2D crystal structure

1. For the crystal lattice and atomic basis vectors for graphene discussed in class:
 - (A) Calculate the x and y positions of the A and B atoms at position $\mathbf{R}=(n\mathbf{a}_1+m\mathbf{a}_2)$ for arbitrary n,m. You will use this for the next problem set.
 - (B) Calculate the area per unit cell in graphene, and the number of atoms per unit area (in $\#/cm^2$). Use this result to calculate the density of a single layer of graphene in units of kg/m^2 .
 - (C) Look up the value of the interlayer spacing in graphite and use this value to calculate the density of graphite. Check this against the standard density of graphite.
 - (D) [PhD students only] Repeat (A-C) above for MoS_2 . For (A), give the x,y,z positions of the single Mo atom and two S atoms. You can find lattice constants for MoS_2 here: <https://materialsproject.org/materials/mp-2815>
2. Reciprocal lattice of graphene:
 - (A) Calculate the area of the reciprocal lattice unit cell for graphene (note that the units are $1/area$)
 - (B) Calculate the locations of the 12 high-symmetry points (6 M points, 3 K points, and 3 K' points) for the 1st Brillouin Zone.

3. Making graphene visible:

Arguably, Andre Geim and Kostya Novoselov won the Nobel prize because of an accident. In some of their early experiments, they exfoliated graphene onto an old piece of silicon, which happened to have about 300 nm of silicon oxide on the surface. This thickness made it easy for them to see the graphene and identify the number of layers, while other research groups trying similar experiments could not see what they were working with. In order to learn why we can so easily see graphene, and how to engineer a system to maximize contrast read the paper by Blake on “Making Graphene Visible.” The paper derives the equations for determining optical contrast from the wave nature of light and using boundary conditions to solve for the magnitude of the reflected light. This is an applied example of a well-known phenomena known as optical interferometry, where light waves confined near a surface can constructively or destructively interfere with each other to locally increase or decrease the amount of light passing through the graphene surface.

Reproduce Figure 3 of the paper.

To complete this assignment, you will need:

- The real and imaginary optical index of graphite, silicon oxide and silicon. Limit the calculation to visible wavelengths.
- Access to your favorite mathematical software (Mathematica, Python, Matlab, Origin lab)

Use the equations in the paper, and the materials properties to calculate the contrast of graphene on silicon oxide versus thickness of the silicon oxide, and versus wavelength of the light. Plot the data in a two-dimensional color plot with substrate thickness on X, wavelength on Y and contrast on Z. Make it pretty, as if you will publish it in a journal (i.e. at the minimum, the axes should be labeled with units, and all text visible. Add a title; an outside reader should be able to understand the data that is presented by looking at your plot). Turn in your code for the calculation and your final plot.